

Shingles, Shakes and Tiles — Materials and Fastening

Your most-missed question in this entire topic is a fastening distance. This lesson is where the numbers live.

YOU HAVE MISSED THIS 6 TIMES

All perimeter concrete roof tiles should be fastened at least ____ inches from both sides of the ridge at the gable rakes and the eaves.

36 inches.

Six times missed, which makes it the most expensive single question you own on the roof. The logic is wind: the edges of a roof — the ridge, the rakes and the eaves — are where wind gets under a tile and lifts it. A three-foot band all the way round the perimeter gets fastened down, and the field in the middle relies on weight and interlock.

The numbers here

Perimeter concrete tile fastening band	36 in
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Shingle nails must penetrate into the sheathing at least	3/4 in
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YOU HAVE MISSED THIS 2 TIMES

Shingles should be attached with nails long enough to penetrate at least _____ into the sheathing.

3/4 inch.

The pieces that sit on top

Nailer	The strip that hip and ridge tiles are installed over. The tiles at the ridge need something to fasten to, because there is no flat deck under them.
Felt interlayment	The strip of felt specified between courses of wood shake roof coverings. Shakes are split rather than sawn, so their surfaces are irregular and water can travel sideways between them.
Lap cement	The asphalt-based cement used to bond shingles together. The same family of material bedded under flashing is roofing cement — lap cement is the name when it is bonding one shingle to another.
Laminated shingle	The heaviest composition shingle per square. It is built from bonded layers, which is where both the weight and the dimensional look come from.
Ceramic granules	The surface of an asphalt shingle. They protect the asphalt from ultraviolet rays, which is what actually ages a shingle roof. Granules in the gutter mean the shingle is losing its sunscreen.
Wood shingles and shakes	Usually made of cedar. A shingle is sawn smooth; a shake is split, so it is thicker and its face is irregular.

YOU HAVE MISSED THIS 3 TIMES

What is specified to be installed between courses of wood shake roof coverings?

Felt interlayment.

YOU HAVE MISSED THIS 2 TIMES

Hip and ridge tiles are installed over a:

Nailer.

A square is a unit of area, not a shingle

One square is 100 square feet of roof. When a question compares shingles "per square" it is comparing weight over the same area, which is why laminated wins.

Say it back

Cover the page. Answer each one out loud.

1. How far in from the ridge, rakes and eaves are perimeter concrete tiles fastened?
2. Say why that band exists rather than just the number.
3. How far must a shingle nail penetrate the sheathing?
4. What do hip and ridge tiles get installed over?
5. What goes between courses of wood shakes, and why do shakes need it?
6. What is lap cement bonding?
7. Which composition shingle is heaviest per square, and what is a square?
8. What do ceramic granules protect the shingle from?
9. What are wood shingles and shakes made of, and how do the two differ?